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Symbiosis Institute of Technology, Pune

*DEPARTMENT OF ELECTRONICS AND
TELECOMMUNICATION*





FOREWORD

The technology field is an exciting place to be right now. Every day brings new problems to solve and new ways to push boundaries. As future engineers, this industry will rely on your fresh perspectives and enthusiasm to drive innovation forward into the future.

Stay curious! The world of electronics and telecom changes so quickly. Believe in yourself. You each have unique talents and a drive to make a difference. Your ideas and creativity will undoubtedly help shape where this industry heads next. Know that success isn't just about achievements or awards. It's also about the positive impact you make. Have passion for what you do. Be determined even when it gets tough. With that motivation guiding you, I know you all have incredibly bright futures ahead. Best of luck in your studies and beyond!

Dr. Prabhat Thakur

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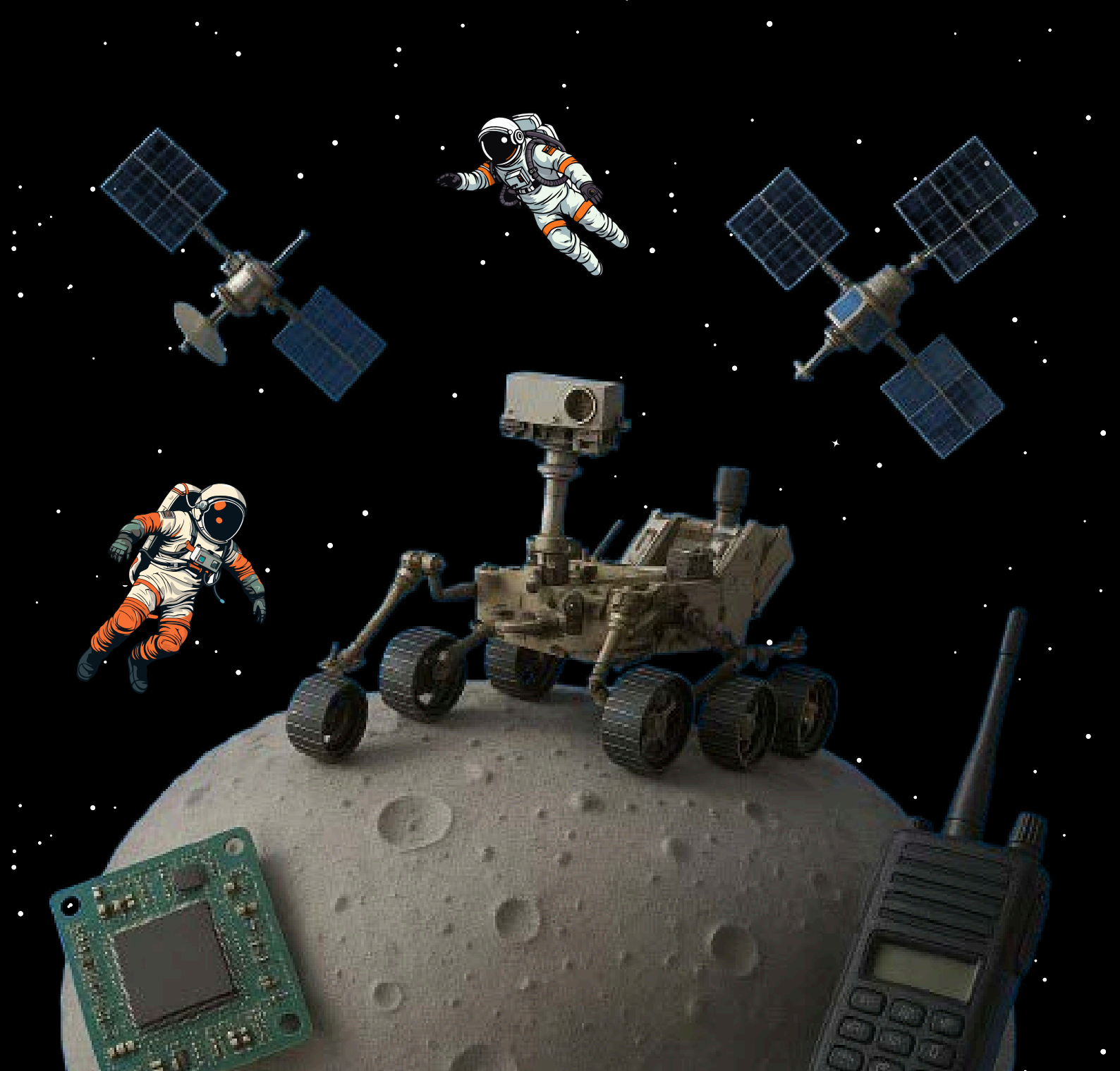
VISION AND MISSION OF THE DEPARTMENT

Department Vision

To emerge as a leading source for Electronics and Telecommunication Engineering, fostering globally proficient engineers to meet the demands of evolving industry and society.

Department Mission

- Foster collaboration with industry to facilitate the acquisition of cutting-edge technologies and contribute to the generation of up-to-date knowledge, enhancing employability and sustainability.**
- Encourage innovation, research, and development, creating an environment conducive to higher education, entrepreneurship, and lifelong learning**
- Cultivate leadership qualities infused with social and ethical values, providing a platform for their development.**



Dear Reader,

Space is, and has always been, endlessly fascinating. Centuries of mankind have devoted countless hours to propel our understanding of the cosmos further, but for now, we continue to orbit around unanswered questions. With the advent of International Space Day 2025, celebrated on 12 April, marking 64 years since the first manned space flight by astronaut Yuri Gagarin in 1961, we bring to you an edition that celebrates humanity's most stellar milestones.

This semester, our department aims to impart cutting-edge insights and insider knowledge of the electronics and intelligence systems that drive space exploration in the digital age. We invite you to join us as we take one small step in our quest to understand the cosmos—but we promise that it's bound to be one giant leap for the



ELECTRONICS IN EXTREME SPACE CONDITIONS

Introduction

Space is the harshest environment known to the mankind. As there is nothing to ease the heat elsewhere and radiation levels that can cook circuits in a nanosecond and temperatures that swing from extreme to worse, there's no place to hand hard served electronics than in space. In contrast to Earth where repairs and replacements are possible, space electronics have to work perfectly – any failure may result in a loss of a billion dollar mission or even human lives. From the first Sputnik satellite to the newest James Webb Space Telescope (JWST), our technology is able to withstand such brutal conditions because of the developments in engineering. Now, with NASA's Artemis program targeting the Moon, ISRO's bold space missions, and commercial space companies like SpaceX expanding horizons, the demand for the less vulnerable and more dependable electronics is ripping than ever.

The Hostile Reality of Space

Electronics created for Earth function in a comparatively stable environment where temperatures are kept constant, weather can be forecasted and keep-up on maintenance can be easily made. However, space is a very different with a much harsher reality. A spacecraft has to withstand extreme shifts in temperature, from -250°C to greater than 250°C in the shadow regions to the sun-facing ones respectively. For instance the Moon's surface warms up in the day and freezes into deep freeze at night. Similarly, spacecraft around earth area as well is subjected to extreme heat from sun and freezing temperatures when earth is in its shadow when in sun. As opposed to the earth, space lacks air to carry heat and thus, conventional methods of cooling such as fans or liquid cooling is of no use. Instead, space-crafts uses radiation and conduction to effectively get rid of heat.

The other big threat to space electronics is radiation. Unlike the Earth, which is protected by the magnetic field and the atmosphere, space is bombarded with the high energy particles from the Sun and the deep space cosmic rays. These particles can damage the microchips, introduce bit flips (Single Event Upsets – SEUs), and degrading of the materials over time. Solar storms and pulses of cosmic radiation can instantly toast vulnerable circuits if strong shielding is not provided for. Shielding of electronic systems from this invisible but destructive force is one of the great engineering problems.

Apart from radiation, another challenge of the vacuum of space exists. In the absence of atmospheric pressure, material used in space electronics has to be carefully chosen to minimize instances of outgassing where trapped gases escape from the material which may contaminate sensitive instruments. Additionally if spacecraft does not have air which is needed for the traditional cooling methods then thermal management becomes essential in design of space crafts.



The trip to space is also dangerous. The start of rockets have severe G-forces, vibrations, and shocks on electronic components. Satellites and spacecraft have to withstand micrometeoroid and space debris impacts, leading to failures or shorting of exposed components. Frictions continue to be an issue in the orbit or on planetary surfaces, thus calling for machines with ruggedized design that can withstand years of operations in these hostile environments. These severe challenges point out why even space agencies and private firms are spending billions in developing and testing electronic appliances for space ventures. An example of failures that this book highlights is the Mars Climate Orbiter that was lost in 1999 due to a simple unit conversion error, or the Hubble Space Telescope, which started off with an optical mistake to victims of catastrophe failures in mission. As we delve further into space, the need for stronger, smarter and more self-healing electronics increases not only for the sheer fact that we need more powerful and intelligent electronics, but so that when future missions must plow through the harshest environments that humanity has ever known, they won't break down.

The process of surviving space is not all about launching a decently stuffed design. It is about being smarter than nature's most merciless forces. Spacecraft electronics have to withstand extreme cold, scorching temperature, and unyielding radiation but without the service of a technician to fix them. Then, how do engineers make the fragile circuits cosmic warriors?



Thermal Management in Space

However, surviving radiation is one thing, while battling is another. Another challenge in space which requires ingenuity is the heat management. Unlike on earth whereby fans and coolants are used to control temperatures, there is no air for heat transmission in space. In order to keep systems running smoothly, spacecraft is dependent on multi-layer insulation (MLI) that resembles golden foil but acts as a thermal barrier. For example, NASA's James Webb Space Telescope has a huge five-layer sun-shield that keeps its instruments at -266°C that is a few degrees above absolute zero. Such extreme thermal protection would make useless its sensitive infrared sensors.

Enduring Launch Forces and Deep Space

Next is the sheer power of going into space. A rocket launch is like an earthquake combined with a hurricane and explosion put together. The vibrations and G-forces generated in lift-off are able to knock parts apart. In order not to fail the mission, engineers create shock-absorbing mounts, reinforced circuit boards, and rugged connectors to stand up to the violent trip into orbit. Deep space probes and satellites also experience another danger – micro-meteoroid hits. Protective shielding and self-healing circuits are embedded to the critical systems in order to counter this. In fact, some sophisticated spacecrafts do have AI diagnostics installed, which sense the damage, and re-route power so as to keep the system going, like a self-curing nervous system.



Radiation Hardened Electronics

Radiation-hardened electronics are some of the most important hazards in space defenses. Normal computer chips might work well back on earth, but one charged particle can flip a bit in memory in the space, random errors possible. To overcome this, space-grade electronics make use of special materials such as silicon-on-insulator (SOI) and gallium arsenide (GaAs) that do not allow radiation-damage to occur. Also, engineers incorporate built-in error correction and redundancy in designers of processors. Thus, if one of them fails, another one picks up automatically. This is the reason why Mars rovers keep functioning even after they have been hit by the cosmic rays for years.

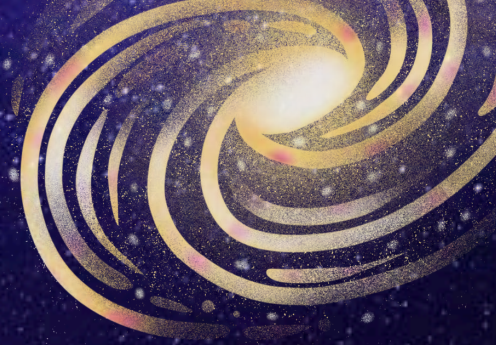
Future of the Space Electronics

Maybe the fastest growing edge of space electronics is the embedding of artificial intelligence. AI-driven fault detection, automated decision-making, and real-time adaptability are growingly used for current space missions. NASA's Perseverance rover, for example, goes on its own way across the surface of the red planet, avoiding obstacles and taking no wait for orders from home. When we plan missions to the Moon, Mars, and beyond, AI-powered spacecrafts will play the core roles in surviving in the unknown worlds. Now, while humanity finds itself at the threshold of interstellar exploration, the electronics that we create have to not only survive but prosper in the most extreme environments. The next generation of space technology won't just endure cosmic radiation, bone-chilling cold, or the violence of rocket launches, but it will learn, adapt, and even repair itself.

Space isn't waiting for us to catch up. The cosmos is vast, relentless, and unforgiving, but it's also filled with infinite possibilities. Our ability to push beyond Earth depends on the innovations we create today. The universe belongs to those bold enough to build the future. The next giant leap won't just be in rockets; in fact it will be in the electronics that carry us into space and the unknown.

BY ABHISAR BEHERA

DATA TRANSMISSION AND TELEMETRY IN SPACE



BY NANDINI BANSAL

For space travel, the transmission and collection of data is essential. It both facilitates communication between satellites and enables interplanetary travel. This technology allows spacecraft to transmit important information to Earth and enables ground control stations to track the missions. Radio frequencies, as well as other methods, allow for the transmission of spaced data.

One of the key principles is Data Encoding: Signals contain data that is encoded. Signals are usually encoded using amplitude modulation (AM), frequency modulation (FM), or phase modulation (PM).

Error Correction and Encoding: Signals deteriorate over time. Methods like Reed-Solomon's and convolutional codes help correct and identify errors.

Signal Propagation: Space signals are transmitted through a vacuum. The signals can be distorted due to the interference of light, radiation from the sun, or Doppler shifts and even from the atmosphere.



Bandwidth and Power Constraints: Spacecraft have limited power and bandwidth, leading to the need for the data compression method to be both efficient and rapid.

Methods of Measurements: Adapting signals cuts down the space for data in a signal. The common already mentioned methods are Amplitude Modulation (AM), Frequency Modulation (FM), and Phase modulation (PM) are common examples.

Telemetry gathers data from remote locations and sends that information to a monitoring center. Telemetry aids in spacecraft health diagnostics as well as scientific research and is crucial for the control of the mission. Reed-Solomon coding and convolutional codes are commonly used for Error Correction and Encoding. Signals do tend to diminish, but these methods are capable of identifying and correcting erroneous signals. Weak signs are able to travel through a vacuum. Atmospheric scattering, solar radiation, and Doppler shifting can interfere with them. Spacecraft are constrained in bandwidth and power, or spacecraft have limited power and bandwidth, resulting in poor transmission and a requirement for data compression.

Distance monitoring systems (telemetry) gather information from remote positions and send it to a control center. In Space, it is needed for mission control, spacecraft health examination, and scientific work.

Key components of space telemetry systems are sensors and instruments that collect data on temperature, pressure, speed, and radiation surrounding the spacecraft. Telemetry Transmitters convert all these sensor readings into electrical signals that are sent Earthward through antennas. Ground Stations receive these telemetry signals, process the sorted data, and return immediate feedback to the mission control center. Data processing units analyze and interpret telemetry data for scientific and operational exploitation.

A few of the aiding technologies to space data transmission and telemetry are RF Communication: The most common way is by using frequency bands like the S-band, X-band, and Ka-band, Optical Communication (Laser-Based Systems): High data rates and bandwidth but requires very precise alignment, Deep Space Network (DSN): This is a global system of antennas for communication between Earth and interplanetary spacecraft, Inter-Satellite Links: The spacecraft also has a similar type of link amongst themselves, which is Telemetry, Tracking, and Command (TT&C). Communication can also be avoided directly through the Earth station.

Challenges in Space Data Transmission

- Signal Attenuation and Lag: As the distance increases, signals become more attenuated, and delay becomes insignificant or deep space missions.
- Interference and Noise: Solar radiation and cosmic background noise can degrade signal quality.
- Solar Only Please: Often, only solar panels or small-scale nuclear are available, limiting the strength of transmission.



Innovations such as the following hold the Future of Space Data Transmission and Telemetry are Zero-Knowledge Protocols: Maintain privacy without revealing your credentials, Artificial Intelligence for Data Processing, Improve telemetry data evaluation for autonomous missions, 5G and Beyond for Space Networks: Building networks with the high-speed, low-latency characteristics we expect of 5G.

Space context: Data transmission and telemetry are vital for space exploration and satellite operations. They provide reliable communication between spacecraft and Earth. Advancements in error correction and communication across space improve mission efficiency and ensure security. This lays the groundwork for future interplanetary exploration and deep-space communication. Next-gen communication systems will require flexible infrastructure that can enable a variety of spacecraft/satellite systems. And this infrastructure needs to evolve with new technologies. National and private space agencies collaborate on system designs and testing new communication protocols and networks. That allows for constant communication with spacecraft all over the solar system.





DEEP DIVE INTO ELECTRONICS IN SPACE ROBOTIC SYSTEM

- SOHAM JADE

Space which was once a distant goal, has never seemed more within reach. The rapid development of autonomous systems and robots enables space exploration at an unprecedented pace and has even transformed the landscape of data collection and processing. We'll explore the role of robotics, and autonomous systems in space exploration in the following text.

Robotics can aid the seemingly sci-fi goal of colonizing space. Now, regardless of one's opinions on the pros and cons of space colonization, we must concede that it has the potential to help mankind's understanding of the world (maybe more so in the case of the global economy, but we digress). Understandably, researchers, politicians, and business executives in particular might find this study lucrative, but make no light of the difficulties of space exploration and colonization. Space exploration and consequently, colonization, comes with significant challenges, such as harsh environments, limited human capacity for long-duration missions, and other complex obstacles. However, advancements in robotics and autonomous systems are rapidly changing what we're able to achieve, making tasks once thought impossible increasingly feasible. Let's dive deep and explore the world of autonomous space systems.

History and Evolution:

The use of robotics, as well as autonomous systems, in the exploration and colonization of space has been a focus of considerable interest for decades. Indeed, a number of studies have been done on this topic, examining a range of aspects of the technology and its many applications as it relates to space exploration and colonization. Speaking in terms of history, the robotics in space, in fact, have been crucial for our comprehension of the universe. The first space mission for which and with which a robot had been used successfully was the Luna 9 of the Soviet Union which landed on the moon in 1966. This has subsequently seen hundreds of robots being used in the studies of planets, asteroids and of comets in space missions. For instance, NASA sent the Mars Exploration Rovers 'Spirit' and 'Opportunity' in 2003 that were to last for 90 days only. Having landed on Mars in 2004, they worked more than a decade in total, gathering a lot of information on geology and Mars' environment. Armed with this background, robotics, and autonomous systems have been pushed in the last few years to build more intelligent and flexible space explorers, and would-be colonists in space. For example, the 2003 launch of the space exploration agency of Europe [the European Space Agency (ESA)'s] Mars Express mission which had a little robot (Beagle2) did not succeed to land on Mars. However, in 2014, a robot called Philae landed on a comet with the help of ESA's mission Rosetta, that has been launched in 2004. Besides, AI and the machine learning are making rapid advances in space exploration and colonization through robotics. For example, the rover Perseverance which belongs to NASA's Mars 2020 mission has Super Cam system. The system guides the rover to find rocks and other interested features on the surface of a planet for examination with the help of the AI and machine learning algorithms.





Enhanced Electronics for Advanced Communication

At its core, a transmitter and receiver are two important elements for any successful space communication. Companies are developing interference-free and optical communication technology in order to provide advanced communication. Thorium Space has launched an ultra-flat scalable matrix antenna that claims to be free of interference from either Earth or space. It uses high radio frequencies along with electronic control of both the transmitting and receiving beams. In April 2023, NASA recorded a significant advancement in space electronics after achieving 200 Giga bits/s throughput on a space-to-ground optical link. This creates a record of the highest data rate attained in optical communications between space and the ground. Laser/optical communications have a number of major benefits ranging from superior efficiency, lighter systems and increased security. The record speed was attributed to TeraByte InfraRed Delivery (TBIRD) system. NASA's Pathfinder Technology Demonstrator 3 (PTD-3) satellite launched TBIRD system to its orbit.

Conclusion:

Human exploration and space living have been transformed by autonomous systems and space robots. From the first automated vehicles like Luna 9 to the more recent AI-operated rovers like Perseverance, robotics-based technologies have demonstrated their capacity to greatly enhance scientific exploration and discovery. Thanks to improvements in communication capabilities like high throughput optical links and interference-free antennas, data transmission to and from space is now more reliable and faster. Strong environmental conditions, high radiation levels, and limited human control from a distance continue to be challenges for both large and small robots. Investigating autonomous systems, novel electronics, and next generation communication systems will continue to be a high priority. As these systems become more advanced, they develop abilities in the exploration of space, while also becoming more capable of enabling space colonization in sustainable manners. This space systems field can offer a vast array of opportunities for the scientists, engineers, and entrepreneurs who are working on developing and securing a future for humankind beyond Earth.

Challenges and Advancements:

- 1) **Autonomy in Harsh Environments:** Independent operation of robots and autonomous systems in hostile and alien environments in terms of navigation, communication and protection is one of the critical difficulties that these systems must overcome in space exploration and colonization. Stronger and more flexible robots that can perform hard tasks with minimal help from human beings are being created to respond to this challenge.
- 2) **Radiation Protection:** Radiation problem is another huge obstacle. The high levels of radiation present in space have the capability to affect electronic devices and severely affect human health. To cope with this problem, scientists are studying new materials and designs for radiation-resistant robots to protect their electronic components.
- 3) **Communication and Control:** The issue of control and communication is one of the most serious problems. It is becoming hard to control and communicate with more autonomous robots. Considering that communication on deep space missions can be delayed for several minutes to hours, this is an issue that is particularly relevant. Scientists are working on new methods of communication such as cognitive radio and laser communication to overcome this hurdle. Such technologies can improve the rate and reliability at which robots can be communicated to.



INTEGRATION OF 5G AND SATELLITE NETWORKS FOR GLOBAL COVERAGE

-Nilay Bhatia

Fifth-generation wireless networking, referred to as simply **5G**, is revolutionizing the world of communication with ultra-high speed data, low latency, and the ability to support a lot of devices at the same time. Such capabilities are needed to enable future technologies such as autonomous automobiles, smart cities, industrial robotics, and virtual or augmented reality. Its biggest shortcoming, though, is that it relies on expensive ground-based infrastructure like fiber-optic cable and dense cell sites. It is financially impossible to install it over highly populated rural ground or geographically difficult areas. A detour in the correct direction has been found by pairing up satellite networks with 5G technology. Satellites have a long-proven history of being in the middle of communication services, where traditional infrastructure is not available.

As demand increases for 5G, satellites are also under consideration as a backbone to offer coverage. Three categories of satellites can be utilized here. These satellites provide low latency and high rates of data. *SpaceX (Starlink)*, *OneWeb*, and *Amazon (Project Kuiper)* have already launched LEO satellite constellations to provide reliable broadband on the entire globe.

Satellites can be integrated into 5G in many different ways.

One of them is **Transparent Integration**, where satellites are simply passive relays to forward data from ground stations to users. Regenerative integration functions on this basis hence it is more efficient and ground infrastructure-independent. The hybrid approach, with a mix of satellite and terrestrial is effective since it offers continuous coverage and performance in any conditions. Some of the latest technologies are making such convergence possible.

Network slicing tops the list since it has the potential to define several virtual networks from shared physical infrastructure. A slice can be allocated to an application. These Satellites employ beamforming and phased-array antennas to offer directed beam pointing and reduction of interference in an effort to improve quality. *Inter-satellite links (ISLs)* enable in-line data transfers between satellites, minimizing the need for constant-on ground station connectivity.



Earth technologies like **Software-Defined Networking (SDN)** and **Network Function Virtualization (NFV)** help in earth-based and satellite system relationships in real time. Satellite-integrated 5G networks are poised to benefit with more than mere scale.

One of the tremendous benefits of providing internet access to remote and disadvantaged communities is that it may close the digital gap by granting access to education, healthcare, and economic opportunities. In air transport and sea transport, constant connectivity can facilitate better navigation, operations, and customer service.. Secure and stable communications networks also assist government and defense ministries operating in hostile regions. It is not all smooth sailing, however.

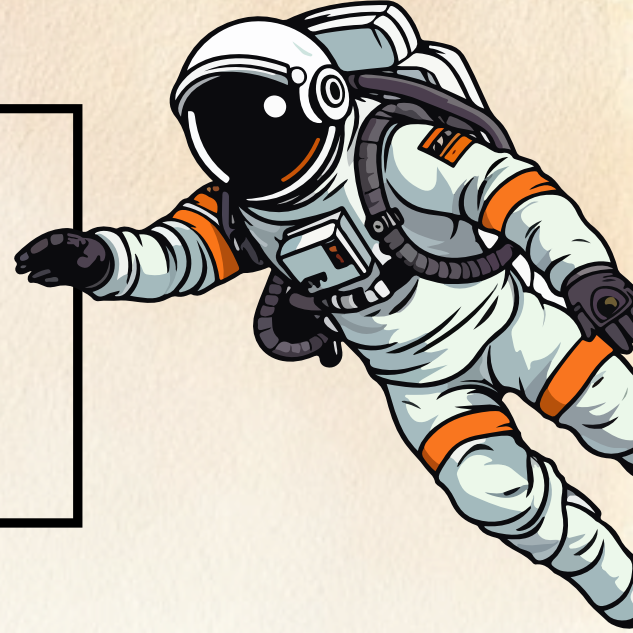
Latency is a problem, that is with *GEO and MEO satellites*, though that is minimized through LEO constellations. **The Doppler effect** due to the high velocity of *LEO satellites* will break signals if not balanced with high-level algorithms. Spectrum management also creates a problem which would require worldwide coordination to enable satellite and ground networks to coexist without interference. The large cost of deployment and upkeep of satellite constellations and ground equipment upgrades is another challenge. Seamless handovers between terrestrial and satellite systems—especially for mobile subscribers—are complex and are being optimized. To fight this adversity, world organizations and standardization organizations are increasing efforts. The *3rd Generation Partnership Project (3GPP)* has included satellite network support—named **Non-Terrestrial Networks (NTNs)**—in its Release 17 and 18 standards. These are implementation standards for the use of satellites in the 5G network. Other organizations like the **International Telecommunication Union (ITU)** and the **European Space Agency (ESA)** are also involved in regulating the spectrum and creating the technology.

In the next few years, satellite convergence with 5G will become the foundation of communications worldwide.

With ongoing development of network optimization via AI, miniaturization of satellite hardware, and cheaper launch opportunities, efficiency and scalability will improve further. With the move towards 6G and later, space-based communications will be part of developing an actual global, smart, and robust infrastructure. *This collision isn't a technology leap—it's an evolutionary imperative to a future in which everyone, anywhere, is connected.*



SENSORS IN SPACECRAFTS



Spacecrafts rely on a wide range of sensors to navigate, monitor their surroundings, and ensure smooth operation. These sensors play a crucial role in space missions, from deep-space exploration to satellite operations and planetary landers. In this article, we will explore the different types of sensors used in a spacecraft, their functions, working principles, applications, and what the future holds for space sensor technology.

Spacecrafts use all sorts of sensors to monitor external conditions, object detection and control of on-board systems. Included amongst the most commonly used common types of sensors are inertial, optical, magnetic, radiation, environmental, proximity, ranging, spectrometers and science payload sensors.

These sensors may be classified broadly as active or passive ones depending on whether they use their own energy or depend on internal sources for energy. The active sensors emit signals and the reflected waves are analyzed while passive sensors detect the naturally emitted or reflected radiation. For instance, proximity sensors and ranging, i.e.: radar, sonar, and lidar, belong to the group of active sensors. On the other hand, inertial and magneto sensors form the group of passive sensors.

Let us talk about each of these sensors briefly:

- **Optical sensors:** These can be active and passive.
 - a) Passive optical sensors including star trackers, sun sensors as well as cameras sense natural light sources for navigation purposes. Star trackers take images of stars and compares them to onboard star maps to determine the accurate location of the spacecraft.. Sun Sensors detect the position of the Sun to assist with attitude control. Cameras are used for planetary observation, docking, and terrain mapping.
 - b) Active optical sensors like Laser Altimeters are used to measure the distance from an orbiting spacecraft to the surface of the planet/asteroid that the spacecraft is orbiting.
- **Magnetic Sensors:** These are passive sensors used for detecting magnetic fields for spacecraft orientation and planetary studies. It includes Magnetometers, which Measure magnetic fields to study planetary magnetospheres or aid in spacecraft orientation relative to Earth's magnetic field.
- **Inertial sensors** are active sensors primarily used for measuring acceleration and angular velocity using gyroscopes and accelerometers.

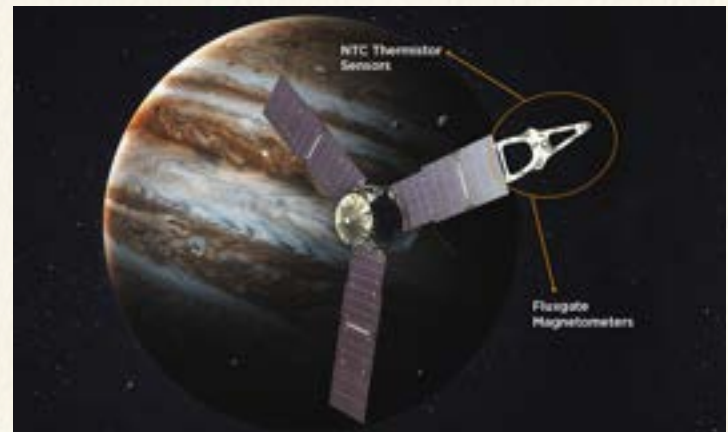


FIG.(A) PASSIVE SENSORS IN A SPACECRAFT

- **Radiation Sensors:** These sensors are passive, and their primary use is to Monitor cosmic rays, solar radiation, and background radiation in space. They play a very important role in spacecraft protection and scientific research.
 - a) Cosmic Ray Detectors: Monitor high-energy particles in space to assess radiation levels.
 - b) Dosimeters: Measure the amount of radiation absorbed by onboard equipment and astronauts.

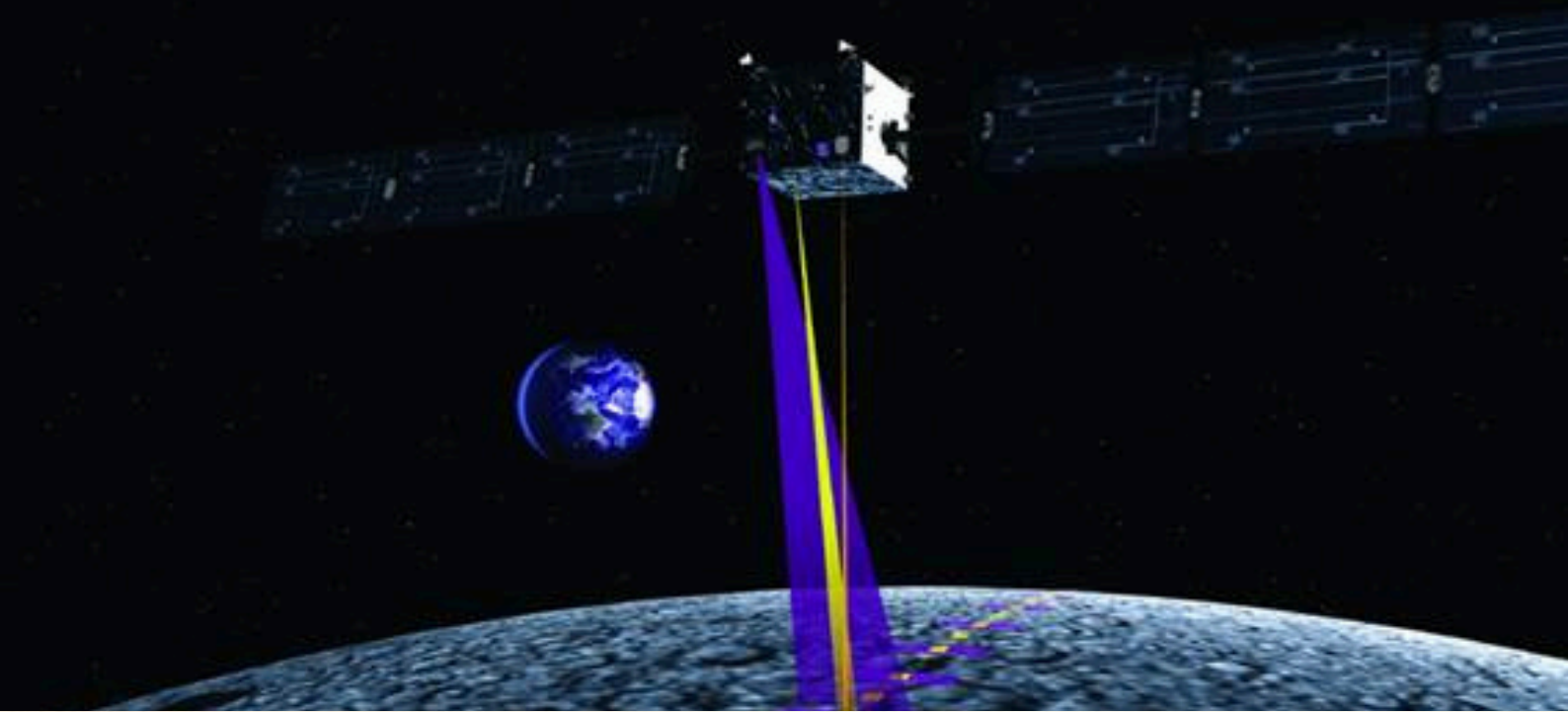
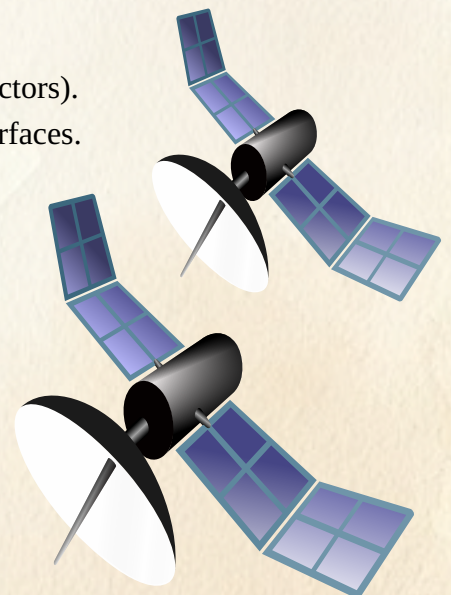


FIG.(B) ACTIVE SENSORS IN A SPACECRAFT

- **Environmental Sensors:** These can also be both active and passive.
 - **a) Passive:** Temperature, pressure, and humidity sensors record conditions without emitting signals. Temperature Sensors monitor thermal conditions to protect spacecraft systems from extreme temperatures. Pressure Sensors measure atmospheric pressure in planetary probes or cabin pressure in crewed missions.
 - **b) Active:** Some environmental sensors (e.g., gas analyzers) actively probe the surroundings.
- **Proximity and Ranging Sensors:** These are active sensors. They include radar, sonar, and Lidar, which emit signals and measure reflections to detect objects. They play a major role in docking, landing, and obstacle avoidance.
 - **a) LIDAR:** Uses laser pulses to determine distance and create 3D maps of planetary surfaces.
 - **b) Radar:** Measures distance and detects objects using radio waves, which are useful for planetary landing and obstacle avoidance.
- **Spectrometers:** These can be both active and passive.
 - **a) Passive:** Optical and infrared spectrometers analyze incoming light to determine composition. For example, Infrared Spectrometers analyze heat signatures to study planetary surfaces and atmospheres
 - **b) Active:** Some spectrometers use lasers or X-rays to excite materials and study emissions. For example, Mass Spectrometers detect and analyze the composition of gases and materials in space.

- **Science Payload Sensors:** These can be both active and passive.
 - **a) Passive:** Collect data from natural emissions (e.g., telescopes, radiation detectors).
 - **b) Active:** Instruments like radar sounders send signals to analyze planetary surfaces.

In short, spacecraft sensors are classified as active or passive, depending on their mode of operation. Passive sensors, such as inertial sensors, magnetometers, and radiation detectors, function by detecting naturally occurring signals without emitting energy. Active-type sensors such as radar, Lidar, and some of the spectrometers, transmit a signal and study its reflections in order to derive data. Categories such as the optical, environmental, and science payload sensors include both the active and passive elements depending on their individual usage. Together, these sensors are of pivotal importance for the navigation of spacecraft, evaluation of the space, exploration of planets, and scientific endeavors, providing accurate data gathering and operations.



APPLICATIONS OF SENSORS IN SPACE MISSIONS:

- **Satellite Operations:** The detection of orbit, communication alignment, and environmental monitoring services are all assisted by sensors.
- **Planetary Exploration:** Rovers and landers rely on sensors for navigations, terrain testing, and research work.
- **Crewed Space Missions:** Environmental and radiation sensors ensure protection of the astronauts as life-supporting conditions are monitored.
- **Asteroid and Comet Exploration:** Sensors are used for determining surface composition and map sites of potential landings.
- **Deep Space Navigation:** Inertial sensors and star trackers are used as navigational tools when spacecraft travel for many interplanetary miles.

Future Developments in Space Sensor Technology:

With the technological development, the sensors of future will be more miniaturized, efficient, and autonomous. AI-combined sensors, miniaturization, quantum sensors, sophisticated imaging systems, and superior radiation convenient are some of the key trends. AI-embedded sensors will facilitate autonomous decision making, and miniaturization will add to the function of CubeSats and small probes. Quantum sensors could provide ultra fine navigation that does not have to rely on GPS, and better imaging systems will deliver higher resolution data for the exploration of planets. In addition, a better radiation shielding will increase the lifetime of the sensors to ensure reliability in extreme space conditions.

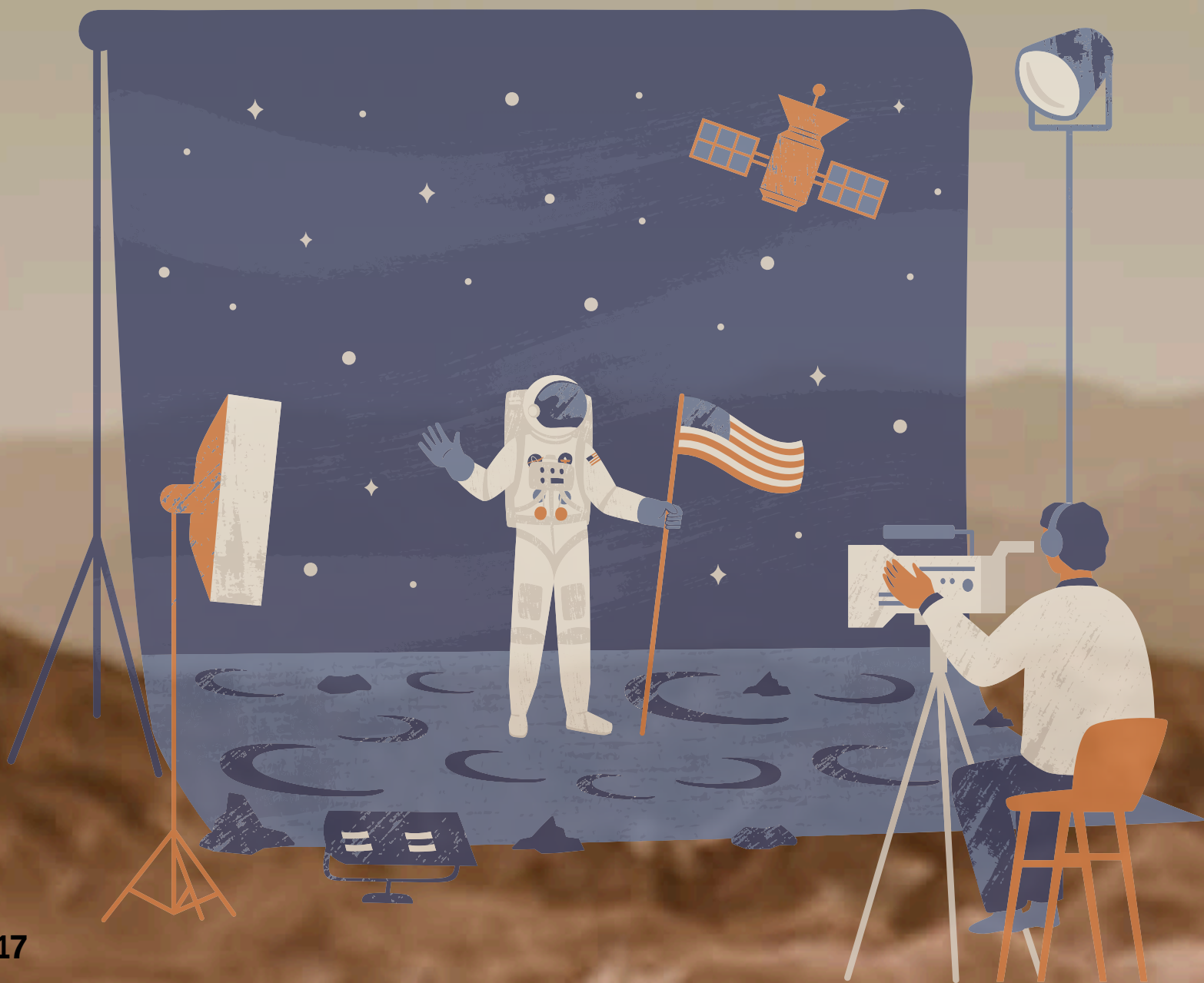
In summing up, Sensors are the very essence of modern space exploration as they guide spacecraft, facilitate scientific findings, and make the missions safe. With the advancement of technology, future sensors will be increasingly utilized, to help remove some asteroids, so it would be easier to navigate the deep space while being more reliable. While every time we make a breakthrough these advancements will help us to get closer to discovering mysteries of the universe, which will confirm this, as sensor technology innovation is a key to humanity's future in space.

~ BY PRIYANSHA GOUR, FY



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SYMBITECH





January at SIT was nothing less than exciting as the college's tech fest, SymbiTech was on the spotlight. Larger, braver, and much more competitive than ever, it united bright minds that were keen on demonstrating their abilities in robotics, coding, AI, finance, and engineering. Stakes were high, the events were intense, and the prizes...? Truly worth the battle.

SOME OF THE EVENTS WERE AS FOLLOWS:

- **Robotics & Mechanical Engineering:** Robo Race saw autonomous robots tackling tricky tracks, while Crane-o-Mania and Tilt'n'Flow challenged participants to build efficient crane systems and explore fluid dynamics.
- **Coding & Cybersecurity:** Debuggers fought bugs in Bug2Bot, while Ctrl+Escape turned coding into a high-stakes escape room. BEYOND THE FIREWALL put cybersecurity skills to the test in an intense defense-and-hacking showdown.
- **AI & Open Source:** The AI Club stole the show with AIBYSS, Alspire 1.0, and a Gen AI Exhibition that displayed jaw-dropping innovations. Meanwhile, the FOSS Club challenged open-source enthusiasts with a tough quiz.
- **Logic, Math & Strategy:** Events like Equation Hi-Li, Techeshi's Castle, and Signal Seekers tested quick thinking, problem-solving, and mathematical expertise.
- **Space & Finance Simulations:** The Martian Marathon pushed space lovers to tackle futuristic challenges, while MONOPOLY and STOCK MOCK put financial strategy to the ultimate test
- **Electronics & Gaming:** Voltage Voyage and Electro Blitz pushed circuit design skills to the limit. Meanwhile, Track Torture blended gaming with simulation, and VentureQuest combined treasure hunts with escape-room elements for an unforgettable thrill.



SymbiTech 2025 was a thrilling ride for tech enthusiasts, and the Electronics & Telecommunications (ENTC) branch made sure to leave its mark. Led by the Next Generation Communication Club (NGCC) and Electronics Design Club (EDC), these events weren't just competitions—they were intense, hands-on experiences that tested skills, teamwork, and quick thinking.

- **Voltage Voyage (EDC):** Classroom 302 turned into a mystery-solving arena where participants raced against time to recover the stolen Voltage Crystal and bring power back to the grid. The event blended logic puzzles, mini-games, and strategy, leading to a nail-biting finale where the smartest minds cracked the case.
- **Electro Blitz (EDC):** Held in the DSP Lab, this fast-paced showdown put electronics skills to the ultimate test. A rapid-fire quiz kicked things off, followed by hands-on circuit challenges. The grand finale? A circuit assembly race where teams scrambled to complete their designs before time ran out.
- **Signal Seekers (NGCC):** A three-stage adventure packed with excitement. Teams decoded clues in the Treasure Hunt, tackled Taboo Quest, where they had to describe concepts without restricted words, and finally, cracked encrypted messages in Morse Code Mayhem. With live leaderboards and exciting prizes, this event kept participants on their toes till the very end.



Reflecting on the success of the fest, the Head of the Electronics Design Club (EDC), Raghav (Third year, E&TC) shared,

"At this year's Techfest, the Electronics Design Club stood out by combining technical expertise with creativity and collaboration. Our flagship events—Voltage Voyage and Electroblitz—offered a unique experience. Voltage Voyage was an exciting treasure hunt with technical and fun, non-technical challenges, while Electroblitz blended a rapid-fire quiz with hands-on circuit-building. What made EDC special was the teamwork and dedication of our core members, who worked tirelessly from ideation to execution. Through these events, we not only showcased the practical side of electronics but also fostered a culture of learning, problem-solving, and collaboration—turning innovative ideas into reality."

Aditi Naringe, head of the Next Generation Communication Club (NGCC), also added,

" The techfest was an incredible showcase of innovation, creativity, and collaboration, bringing together students to explore and celebrate technology. NGCC focused on making topics like 5G and IoT more interactive and engaging, turning learning into an experience rather than just a competition. With challenges that combined problem-solving, teamwork, and hands-on application, Signal Seekers offered something lively and hands-on. Seeing teams crack codes, hunt for components, and build working circuits was a reminder of how learning is at its best when it's immersive and fun."

Together, these events captured the true spirit of SymbiTech 2025—where engineering met strategy and technical skills turned into thrilling challenges. It wasn't just about competition; it was about thinking on your feet, solving real-world problems, and embracing the rush of innovation. Participants walked away not only with newfound skills and experiences but also with friendships, memories, and the sheer thrill of pushing their limits.

5G LAB INAUGURATION BY DEPARTMENT OF ENTC

Introduction:

Towards the end of March 2025, the department of electronics and telecommunication announced the inauguration of 5G Use Case Lab, a cutting-edge facility dedicated to exploring and demonstrating the transformative potential of 5G technology.

This visionary initiative towards a better future and resourceful learning for students was sponsored by the **Department of Telecommunications (DOT), Govt. of India** and had been under construction since the past 6 months. TCIL and VVDN Technologies also played a vital role in bringing this project to life.

During the inauguration, the event witnessed the presence of many of the esteemed experts from leading telecom organizations, to name a few, **Rajesh Kanungo, DDG (Compliance), DoT, Govt. of India, Dr. Sujata Dutta, Senior Vice President, Technology, Vodafone Group, Munir Sayyad, Vice President, Reliance JIO, Yogesh Shah, Vice President, Automation Edge , Deep Gupta, Director, VVDN Technologies, Prasanna Deshpande, MathWorks,** And many others, whose insights added immense value to the event.

The event also showcased the exceptional talent of our students, who demonstrated innovative projects powered by 5G technology. Their creativity and technical expertise were truly commendable. Additionally, industry leaders shared their invaluable insights on the critical relationship between Industry and Academia and explored the diverse dimensions of the 5G Use Case Lab, emphasizing its role in driving cutting-edge solutions.

This lab is a hub for innovation, where we'll develop and test real-world 5G applications across industries, including healthcare, manufacturing, smart cities, and entertainment. By leveraging ultra-fast speeds, low latency, and high connectivity, we aim to unlock new opportunities and drive digital transformation on a global scale.

This initiative is a significant step towards driving innovation and exploring the vast potential of 5G technology. It's inspiring to see the collaborative efforts between academia and industry in shaping the future of connectivity.



INDUSTRY CONCLAVE 2025



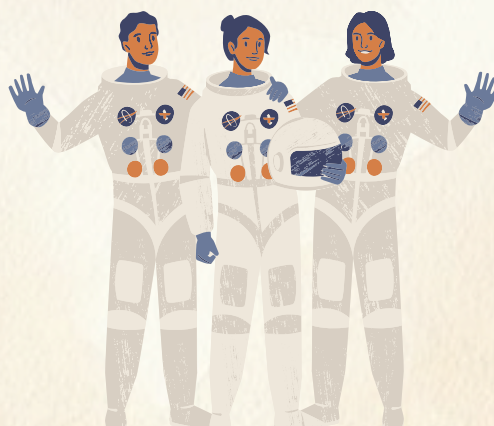
Introduction:

Industry Conclave 2025 was organized at Department of Electronics and Tele-Communication SIT, Symbiosis Institute of Technology, Pune, and it was a much needed learning experience. This energetic mingling of students, faculty and industry leaders helped facilitate informative discussions about bridging the gap between what business needs and universities have to offer.



The conclave was a good platform for networking, sharing of information and cooperation in future, the participants said. To ensure that students are able to get their hands on industry-led innovations, experts discussed new trends.

And thanks for the faculty, students, and panelists who made this the best possible use of their time. The success of such programs offers hope for in the future of engineering education that emphasizes the importance of sustained engagement within academic groups.



Industry Visit



On 11th March 2025, the 2nd year electronics and telecommunication students participated in an industrial visit to Minilec India Pvt. Ltd known for manufacturing and exporting of industrial electronic products. The company was founded in the year 1965 by Yashavant G. Ghaisas as part of Yash Prabha group. Starting operations as a manufacturer of single phasing preventers (phase failure relays), Minilec is today a leading group involved in 100% indigenous production, marketing, and export of reliable industrial electronic products for protection, control, and monitoring.

Students traveled to the facility by bus and, upon arriving at the location, were introduced to Mr. Swapnil Bhagare, the Executive – HR and Admin, who provided a brief introduction into their working and the type of projects the company works on and then led the tour into their workplace. The students were divided into two groups and the visit was also divided into 4 parts.

First, the students were introduced to the Phase failure relays (PFR) and monitoring relays (MR), both of which are manufactured and tested there. PFR which are suitable for monitoring incoming 3-phase supply from Mains (Electricity distribution line). Being independent of load, they can be used for any HP/KW rating of load and Monitoring relays which are best suitable for monitoring balanced or unbalanced supplies, either of single-phase, 3-phase 3-wire or 3-phase 4-wire loads powered by generators, UPS, stabilizers, inverters, or Mains (by Electricity Boards / Utilities); in AMF panels, PCCs, distribution boards and for power monitoring of individual loads/motors / pumps. The second part of the visit was manufacturing Printed circuit boards (PCB). PCBs are used for electrical connectivity and mechanical support. There are different types of PCBs like Single side, Double side, Multi-layer – Rigid, Flexible. The third part were the students learned about Time management which included topics like organizing and planning of the work. Lastly, they talked about how moral and ethics are important at the work place where topics like punctuality, responsibility, respect and team management were talked about.



The session was very interactive and engaged both the presenters and the students in an active manner. It had a great balance of practical knowledge and theoretical knowledge about PCBs and relays that sparked the curiosity of the youth in attendance. The discussion encouraged students to ask questions and share their thoughts. Throughout the session, Mr. Bhagare and his colleagues provided the students with valuable insights into emerging trends in industry, career pathways in a number of fields, and the implications of the technologies they discussed. Their expertise was a good source of inspiration and motivation for the students.

VISIT TO IITB NANO FABRICATION LAB

Date of Visit: April 5th, 2025

Purpose of Visit: Educational Visit



Introduction:

On April 5th, 2025, an educational visit was conducted at the IIT Bombay (IITB) Nanofabrication Lab. The purpose of the visit was to familiarize participants which were a total of 15 students and 10 from 3rd year EnTC and 5 from 2nd year having 2 members of faculty Dr. Sankit Kassa and Dr.Mousami Bhanja to guide us along the visit with the various advanced tools and equipment used in nanofabrication and device characterization. The visit took us through multiple laboratories within the facility, including the Micro 1 Lab, Micro 2 Lab, Nano Lab, Biosensors Lab, and the Device Characterization Lab and more.

Visit Overview:

1. Micro 1 Lab: The Micro 1 Lab is equipped with several tools essential for fabrication and material processing. The following equipment was observed:

- **2-inch Boron Solid Source Diffusion Furnace:** This furnace is used for the diffusion process in semiconductor manufacturing, providing high-quality material integration for various devices.
- **2-inch CMOS-Specific FGA Furnace:** A specialized furnace designed for the fabrication of CMOS (Complementary Metal-Oxide-Semiconductor) devices, ensuring precise material processing.
- **2-inch Dry Oxidation Furnace:** Utilized to grow a high-quality oxide layer on silicon wafers, which is crucial for microelectronic applications.
- **Parylene CVD SCS (Chemical Vapor Deposition System):** Used for depositing parylene thin films, which are often employed as insulative coatings in devices.

- **Gold Electroplating System:** This system is used for electroplating thin gold layers on semiconductor substrates, which is important in microelectronics for interconnections.

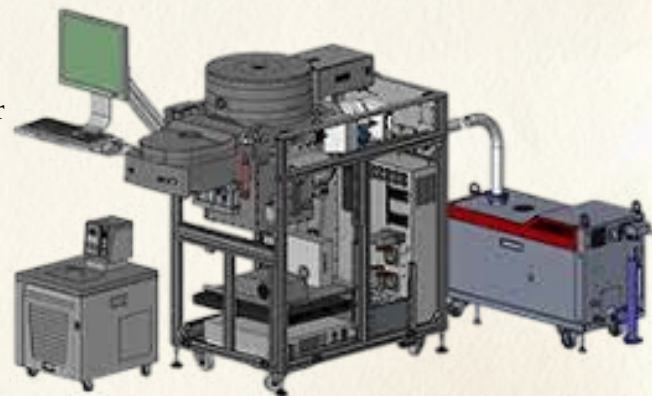
2. Micro 2 Lab: The Micro 2 Lab showcased tools essential for advanced thin film deposition and material analysis:

- **Hot Wire CVD (HWCVD):** A method used for thin film deposition of materials like silicon and carbon, typically used for applications in solar cells and semiconductor industries.
- **Plasma Immersion Ion Implantation (PIII):** This tool enables the implantation of ions into substrates using a plasma field, which is critical for modifying material properties at the atomic level.
- **4-Probe Measurement (Automatic):** A highly precise tool used for measuring the electrical properties of materials, particularly resistivity, which is key in semiconductor research.
- **X-ray Photoelectron Spectroscopy (XPS):** A surface-sensitive technique used for analyzing the chemical composition of materials, particularly useful in materials science and nanotechnology.



3. Nano Lab: In the Nano Lab, several tools designed for nanoscale fabrication and characterization were demonstrated:

- **Inductively Coupled Plasma CVD (ICP CVD):** A tool for high-quality deposition of thin films using plasma, commonly used in semiconductor fabrication.
- **Olympus Microscope (Nano Lab):** A high-resolution microscope used for visualizing nanostructures and thin films with extreme precision.
- **Rapid Thermal Processing (RTP):** This system is used for the rapid heating and cooling of wafers, which is essential in semiconductor processing for fast annealing of thin films.
- **Ultech Furnace (Dry Oxide):** A furnace used for growing oxide layers on silicon wafers for use in semiconductor devices.
- **Sputter ATC 2200:** A sputtering system used for thin-film deposition, which is a crucial technique in the fabrication of microelectronic devices.
- **Nano Wet Bench:** A wet-processing tool used for cleaning and etching materials at the nanoscale, important for preparing substrates in microelectronics.



4. Biosensors Lab (7th Floor): The Biosensors Lab focuses on the development of biosensors and related technologies. Tools observed include:

- **FTIR (Fourier Transform Infrared Spectroscopy):** A tool used for material characterization by analyzing the absorption of infrared light, useful in studying organic compounds and biological samples.



- Potentiostat: A device used for measuring electrochemical properties, particularly in the development and testing of biosensors.
- Fluorescence Microscope (Z1): A microscope that uses fluorescence to visualize biological and nanostructural samples, which is useful in biosensor research and development.
- Laser Engraving System (LES): A high-precision tool used for engraving designs onto materials, particularly in biosensor applications.

5. Device Characterization Lab (NanoE Building, 3rd Floor): The Device Characterization Lab focuses on testing and characterizing various electronic devices. The following tools were showcased:

- Polaris (Stress/Long Time Measurements): Used to measure mechanical stress and reliability in materials and devices over time, crucial in long-term device performance analysis.
- Proxima (Fast IV Measurement/B1500): This system measures the current-voltage (IV) characteristics of devices quickly, helping to analyze their electrical properties.

Key Learnings and Insights:

The visit provided a comprehensive overview of the state-of-the-art tools used in the fields of nanofabrication, materials science, and device characterization. Each tool demonstrated its role in advancing micro and nanoelectronics, as well as its specific application in the fabrication of high-performance devices. Key takeaways include:

- The importance of precise material processing and characterization in the development of semiconductor devices.
- The role of nanofabrication techniques in driving innovation in fields such as biosensing, semiconductor electronics, and nanotechnology.
- The cutting-edge technology and expertise available at IITB, which positions it as a leader in nanofabrication research and development.

Conclusion:

The educational tour of the IITB Nanofabrication Lab was a valuable experience that deepened our understanding of advanced fabrication and characterization techniques. The exposure to the various tools and equipment in the lab provided insight into their critical roles in research and development within the field of nanotechnology. This visit has inspired further interest in nanofabrication processes and their applications in both academia and industry.



RELIANCE JIO LAB AND L&D CENTRE, Navi Mumbai

As a part of Industry and Academic collaboration, students of second and third year were taken to visit the *Reliance Jio Lab and Learning & Development Centre in Navi Mumbai*. Our faculty members, *Dr. Prabhat Thakur and Dr. Mangal Singh* were guiding us through the visit. It was aimed at enabling us to understand the applications of 5G technologies in real life and its potential to transform industries. We also know us through the visit. It was aimed at enabling us to understand the applications of 5G technologies in real life and its potential to transform industries. We also know that the industry presently is working on 6G too. This visit helped fuel the fire for the students to see a future in the telecommunications industry.

About the Jio fiber AND Jio Air-fiber:

The visit started with an introduction to two main technologies designed to provide high-speed internet to urban and rural India. Jio Fiber offers fast, stable fiber optic connectivity with low latency, which is crucial for *supporting modern communications, streaming platforms, and smart-home technologies*.

Jio Air-fiber on the other hand provides a *wireless 5G connection* where laying any fiber infrastructure is challenging and this technology enables new possibilities for internet access in rural areas, remote education, and telehealth. These services are not just providing the internet at higher speeds but the infrastructure for a *smarter and more connected nation*.

Real-World Use Cases That Inspired Us :-

- **Automated Mobility:** We were showcased how smarter and real-time communication between the vehicle and the user takes place via an app called **Jio Humsafar**. The platform used various monitoring systems like cameras, speed monitors, and an AI tool that monitors driver behavior and in the event of abnormality, the system could alert the user. We were also showcased *Jio's own battery charging system for EVs*.
- **Home Automation:** We saw how smart homes are connected using Jio fiber and air-fiber and how easy it is for the user to control domestic equipment like lights , appliances, security systems, etc. *This ease of IoT innovation and integration gives users access at home, help for disabled people and enable energy efficiency*.
- **Healthcare:** The healthcare demonstration focused on telemedicine and real-time health monitoring, and its fundamental change on patient care. **Sensors and connected medical devices** now allow monitoring health conditions constantly, tracking blood pressure, heart rate, and oxygen levels to be sent in real-time to health service providers so that health

- service providers can respond with timely interventions and patients can have remote consultations. Patients are now having more access to healthcare even in *rural areas*.
- **Environment Monitoring:** We were showcased the **Jio True 5G drones** that use AI to detect the hotspots for pollution, and provide live feedback to the engineers that monitor it remotely. They were used to collect garbage from various places including the river Ganga. *These could carry about 20 Kg of waste.* These are being currently used by the government of India across various cities for the **clean India mission**.
- **Education:** Jio has an online education platform called **Perpetual**. This platform uses virtual classrooms and *open edX platform to teach in various government schools in rural and urban India*. This also uses AR/VR in education to enable the students to have an immersive experience.
- **Agriculture:** We learned about smart *farming practices* supported by IoT and connectivity solutions. Sensors are used in farming fields that enable real time monitoring of *soil moisture conditions, temperature, and nutrient levels*. This information helps in automating irrigation and giving farmers insight into best farming practices which is *increasing crop yield, utilizing fewer resources, and decreasing agriculture's environmental impact*.

Live demonstrations with Jio experts provided ways in which these technologies are already shaping life experiences in our rural and urban communities.

Fueling Innovation and Research:

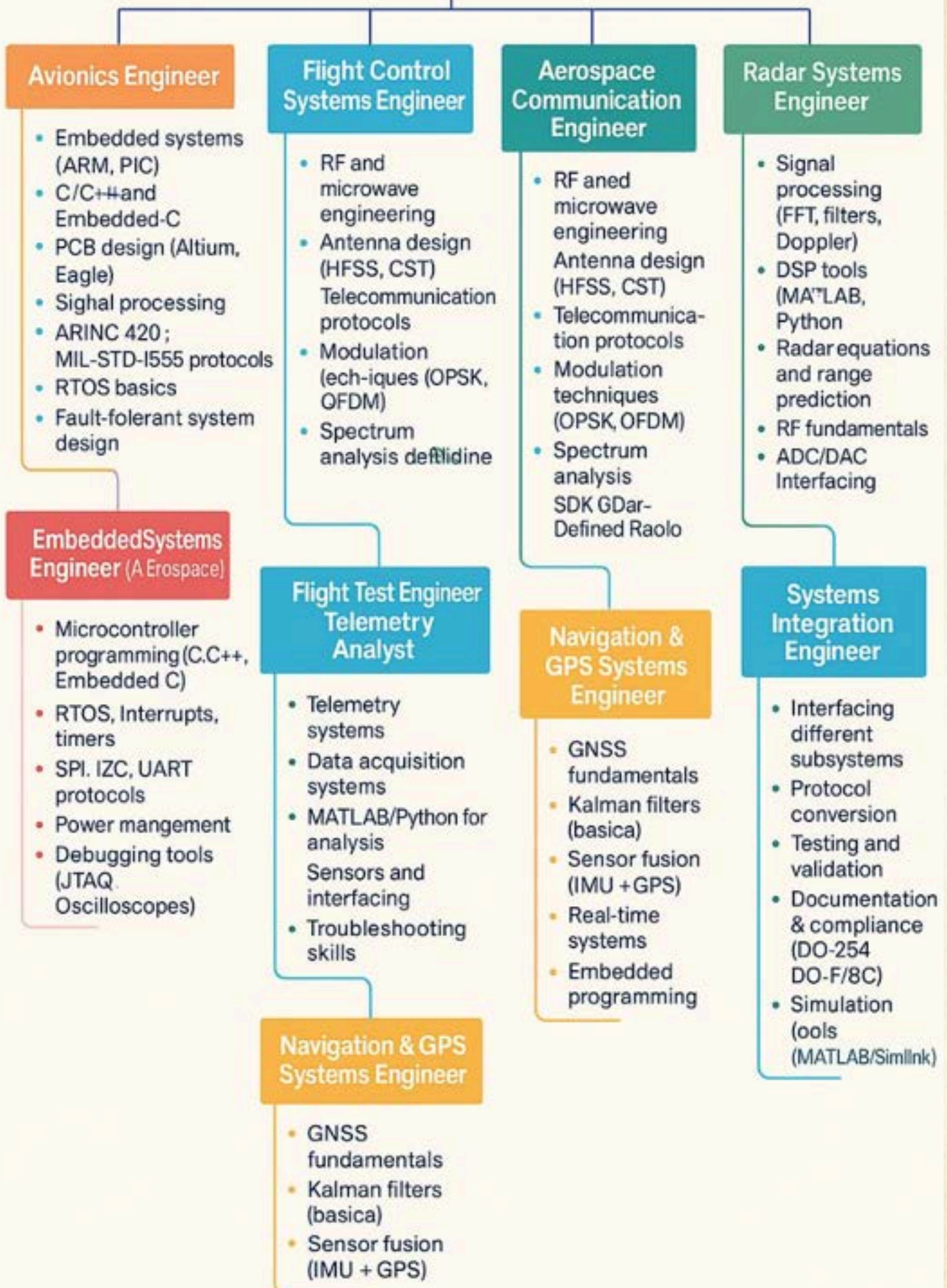
The tour also complements the **5G Use Case Lab project at SIT**, where students can research, develop and test their ideas as prototype technologies. These next generation systems will really mature in the not-too-distant future. *For students like us, the tour was not simply an industry visit, it was an experience that found inspiration, illuminated our curiosity for research and innovation, and introduced reality to what we learn in class.*

Looking Ahead:

With the world moving into *5G, 6G and beyond*, this visit reiterated the importance of being informed, curious, and engaged with technological development. As future engineers, we felt more driven than ever to understand how we can be part of the *innovation wave in India and the world*.

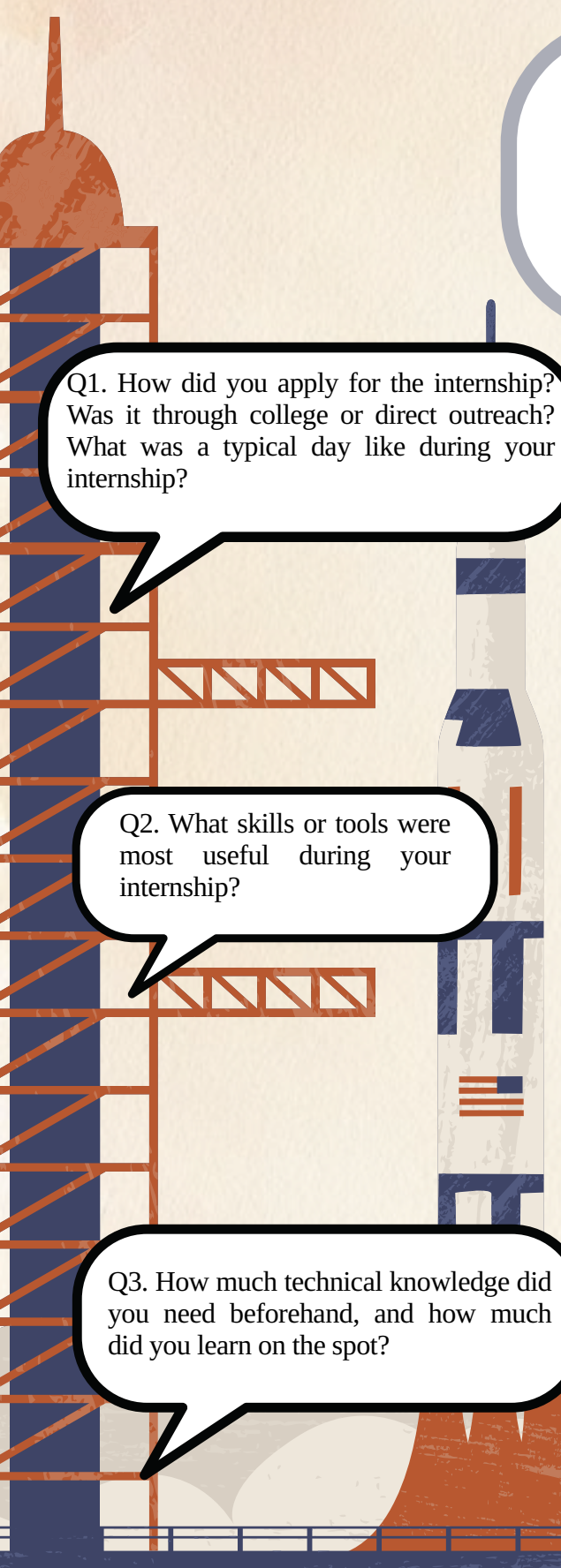


Aeronautics Job Roles for ENTC Engineers



STUDENT SPOTLIGHT

Interview with ISRO Intern Aman Ballabh, SIT directed by our very own team member, Priyansha Gour.

A stylized illustration of a rocket launch. On the left, a blue and orange rocket is shown in profile, with orange flames at its base. To its right, a white rocket with blue and orange accents is shown in profile, also with orange flames at its base. The background is a light beige color with faint, stylized clouds or smoke patterns. The entire illustration is framed by a dark blue border at the bottom.

Aman Ballabh is a final-year Robotics Engineering student at Symbiosis Institute of Technology, currently interning at ISRO, Ahmedabad. He is passionate about Robotics, Machine Learning, and Computer Vision, with a strong focus on developing impactful technological solutions. Aman actively explores complex engineering challenges and aims to contribute to the advancement of robotics through innovation and research.

Q1. How did you apply for the internship? Was it through college or direct outreach? What was a typical day like during your internship?

“I got this internship with the help of our college. The work hours weren’t fixed—some days it could stretch from morning till late at night, depending on the task. Deadlines were tight, so holidays were pretty rare. They provided lunch and dinner on-site too, and it was quite affordable—just ₹15 for a fixed thali.”

Q2. What skills or tools were most useful during your internship?

My background in robotics played a big role. We were building actuators from the ground up—assembling everything from gears to motors. It had to be done with incredible precision, achieving resolutions under 20 nanometers. So, hands-on skills and attention to detail were key.

Q3. How much technical knowledge did you need beforehand, and how much did you learn on the spot?

“For my role, a strong foundation in mechanical engineering, electronics, and computer vision was essential. That said, the required knowledge can vary a lot depending on the specific project. My advice would be to keep sharpening your skills early on so you’re ready to handle advanced-level work when the time comes.”

Q4. What should students start learning or working on if they want an internship at ISRO?

“Not really anything special but CGPA, all I can see were ppl in the league of 8.5+cgpas, and for skills, you should just work on yourself and apply for internship they will see your skills and match with their preference”

Q5. Can you describe the project you worked on at ISRO?

“I can try to explain but it is confidential so I won't be able to describe it exactly. There is a hexagonal shaped structure which is combined to arrange in such positions which resembles a telescope. These hexagons which are the mirrors are then controlled to focus the beams at the detector which converts the rays to give us an spectacular image of the cosmic world”



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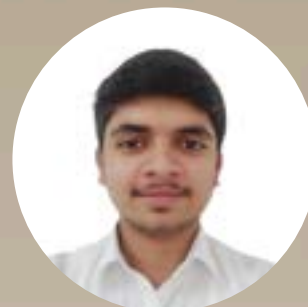
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